

Mutual Fund Application

Stocks	A	B	C
Shares owned	50,000	100,000	200,000
Price @ 10:00	105	73	54
Price @ 4:30	110	78	52

Fund Shares : 500,000

Fund Liabilities = \$200,000

[Q1]: Buy 100 shares in fund at 10 AM on 09/02
Need the fund's NAV

$$NAV = \frac{\text{Assets} - \text{Liabilities}}{\# \text{ shares of fund}}$$

NAV is always calculated

at the end of the trading
day (closing)

Value of Assets =

$$\begin{aligned} & 50,000 \times 110 + \\ & 100,000 \times 78 + \\ & 200,000 \times 52 \end{aligned}$$

\$ 23.7 million

$$\text{NAV} = \frac{23.7 \text{ mil} - \$200,000}{500,000}$$

$$\boxed{\text{NAV} = \$47}$$

Cost of buying 100 shares:

$$\$47 \times 100 = \$4,700$$

Q2

Investor wants to
redeem 10,000 shares
from fund.

New NAV?

cash value
of 10,000 shares
↑

$$\text{New NAV} = \frac{23.7 \text{ mil} - 200,000 - 47 \times 10,000}{500,000 - 10,000}$$

$$\boxed{\underline{\text{NAV}} = \$47}$$

Mutual Fund Taxation

	Bought	Sold
<u>AAPL</u> :	\$150	\$230

Size of Position	300,000	100,000
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$$\text{Capital gain: } (230 - 150) \times 200,000 = 16,000,000$$

300,000 - 100,000

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\$ 16 mil

$$\text{Tax: } 15\% \times \$16 \text{ mil} = \$2.4 \text{ mil}$$

$$\text{Ownership share in the fund} = \frac{10,000}{500,000} = 2\%$$

$$\text{Your tax bill: } 2\% \times \$2.4 \text{ mil} = \$48,000$$

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